complete java class code example of each combination of intermediate and terminal operations

filter() boolean checks + count() no custom logic allowed

filter() null checks + count() no custom logic allowed

filter() comparisons + count() no custom logic allowed

filter() method calls returning boolean + count() no custom logic allowed

distinct() rely on equals() / hashcode() + count() no custom logic allowed

distinct() immutable objects + count() no custom logic allowed

map() transform objects + count() no custom logic allowed

map() create new objects + count() no custom logic allowed

map() call pure functions + count() no custom logic allowed

flatmap() return stream + count() no custom logic allowed

flatmap() split data + count() no custom logic allowed

flatmap() expand collections + count() no custom logic allowed

sorted() natural ordering + count() no custom logic allowed

sorted() custom comparator + count() no custom logic allowed

sorted() multi-field compare + count() no custom logic allowed

peek() logging + count() no custom logic allowed

peek() debugging + count() no custom logic allowed

peek() metrics + count() no custom logic allowed

Below are **all 18 programs converted to Java 1.5 compatible code**.

Since Java 1.5:

- ❌ No Stream API
- ❌ No Optional
- ❌ No lambda expressions
- ❌ No method references
- ❌ No List.of(), Map.of()
- ❌ No Objects.hash()
- ❌ No Comparator.comparing()

We replace everything with:

- ✔ for-each loops
- ✔ Iterator
- ✔ Collections / Arrays
- ✔ HashSet for distinct
- ✔ Manual sorting with Comparator
- ✔ Null checks
- ✔ Primitive counters instead of AtomicInteger

Below are **complete, standalone Java class examples** for **each requested Stream combination**, where:

- **Terminal operation = count()**
- **No custom aggregation / reduce / collect logic**
- Only **built-in Stream operations**
- Java **8+ compatible**
- One clear intent per class

✅ "No custom logic allowed" respected:
count() is the **only terminal**, no reducers, no collectors, no mutation.

---

1 filter() boolean checks + count()

```
import java.util.*;

public class FilterBooleanCount {
   public static void main(String[] args) {
      List<Integer> numbers = new ArrayList<Integer>();
      numbers.add(10);
      numbers.add(15);
      numbers.add(20);
      numbers.add(25);

      long count = 0;

      for (Integer n : numbers) {
         if (n % 2 == 0) {
            count++;
         }
      }
      System.out.println(count);
   }
}
```

---

2 filter() null checks + count()

```
import java.util.*;

public class FilterNullCount {
   public static void main(String[] args) {
      List<String> names = Arrays.asList("Anurag", null, "Amit");

      long count = 0;

      for (String name : names) {
         if (name != null) {
            count++;
         }
      }

      System.out.println(count);
   }
}
```

---

3 filter() comparisons + count()

```
import java.util.*;

public class FilterComparisonCount {
   public static void main(String[] args) {
      List<Integer> salaries = new ArrayList<Integer>();
      salaries.add(20000);
      salaries.add(40000);
      salaries.add(60000);

      long count = 0;

      for (Integer s : salaries) {
         if (s >= 30000) {
            count++;
         }
      }

      System.out.println(count);
   }
}
```

---

4 filter() method call returning boolean + count()

```
import java.util.*;

public class FilterMethodCount {

   static class User {
      boolean active;
      User(boolean active) { this.active = active; }
      boolean isActive() { return active; }
   }

   public static void main(String[] args) {
      List<User> users = new ArrayList<User>();
      users.add(new User(true));
      users.add(new User(false));
      users.add(new User(true));

      long count = 0;

      for (User user : users) {
         if (user.isActive()) {
            count++;
         }
      }

      System.out.println(count);
   }
}
```

---

5 distinct() rely on equals() / hashCode() + count()

```
import java.util.*;

public class DistinctEqualsCount {

   static class Person {
      String name;

      Person(String name) { this.name = name; }

      public boolean equals(Object o) {
         if (this == o) return true;
         if (!(o instanceof Person)) return false;

         Person p = (Person) o;
         return name.equals(p.name);
      }

      public int hashCode() {
         return name.hashCode();
      }
   }

   public static void main(String[] args) {
      List<Person> people = new ArrayList<Person>();
      people.add(new Person("Anurag"));
      people.add(new Person("Anurag"));
      people.add(new Person("Ravi"));

      Set<Person> unique = new HashSet<Person>(people);

      System.out.println(unique.size());
   }
}
```

---

6 distinct() immutable objects + count()

```
import java.util.*;

public class DistinctImmutableCount {
   public static void main(String[] args) {
      List<String> values = new ArrayList<String>();
      values.add("Java");
      values.add("Java");
      values.add("Python");

      Set<String> unique = new HashSet<String>(values);
      System.out.println(unique.size());
   }
}
```

---

7 map() transform objects + count()

```
import java.util.*;

public class MapTransformCount {
   public static void main(String[] args) {
      List<String> names = new ArrayList<String>();
      names.add("anurag");
      names.add("amit");

      long count = 0;

      for (String name : names) {
         String upper = name.toUpperCase();
         count++;
      }

      System.out.println(count);
   }
}
```

---

8 map() create new objects + count()

```
import java.util.*;

public class MapCreateObjectCount {

   static class UserDTO {
      String name;
      UserDTO(String name) { this.name = name; }
   }

   public static void main(String[] args) {
      List<String> names = new ArrayList<String>();
      names.add("Anurag");
      names.add("Amit");

      long count = 0;

      for (String name : names) {
         UserDTO dto = new UserDTO(name);
         count++;
      }

      System.out.println(count);
   }
}
```

---

9 map() call pure functions + count()

```
import java.util.*;

public class MapPureFunctionCount {

   static int square(int n) { return n * n; }

   public static void main(String[] args) {
      List<Integer> numbers = new ArrayList<Integer>();
      numbers.add(2);
      numbers.add(3);
      numbers.add(4);

      long count = 0;

      for (Integer n : numbers) {
         int result = square(n);
         count++;
      }

      System.out.println(count);
   }
}
```

---

10 flatMap() return Stream + count()

```
import java.util.*;

public class FlatMapStreamCount {
   public static void main(String[] args) {

      List<List<Integer>> lists = new ArrayList<List<Integer>>();

      lists.add(Arrays.asList(1, 2));
      lists.add(Arrays.asList(3, 4));

      long count = 0;

      for (List<Integer> list : lists) {
         for (Integer value : list) {
            count++;
         }
      }

      System.out.println(count);
```

```
    }
}
```

---

11 flatMap() split data + count()

```
import java.util.*;

public class FlatMapSplitCount {
    public static void main(String[] args) {

        List<String> lines = new ArrayList<String>();
        lines.add("Java Stream");
        lines.add("Flat Map");

        long count = 0;

        for (String line : lines) {
            String[] words = line.split(" ");
            for (int i = 0; i < words.length; i++) {
                count++;
            }
        }

        System.out.println(count);
    }
}
```

---

12 flatMap() expand collections + count()

```
import java.util.*;

public class FlatMapExpandCount {
   public static void main(String[] args) {

      Map<String, List<String>> data = new HashMap<String, List<String>>();
      data.put("A", Arrays.asList("1", "2"));
      data.put("B", Arrays.asList("3"));

      long count = 0;

      for (List<String> list : data.values()) {
         for (String value : list) {
            count++;
         }
      }

      System.out.println(count);
   }
}
```

---

13 sorted() natural ordering + count()

```
import java.util.*;

public class SortedNaturalCount {
   public static void main(String[] args) {

      List<Integer> numbers = new ArrayList<Integer>();
      numbers.add(5);
      numbers.add(1);
      numbers.add(3);

      Collections.sort(numbers);

      System.out.println(numbers.size());
   }
}
```

---

14 sorted() custom comparator + count()

```
import java.util.*;

public class SortedCustomCount {
   public static void main(String[] args) {

      List<String> names = new ArrayList<String>();
      names.add("Anurag");
      names.add("Ravi");
      names.add("Amit");

      Collections.sort(names, Collections.reverseOrder());

      System.out.println(names.size());
   }
}
```

---

15 sorted() multi-field compare + count()

```
import java.util.*;

public class SortedMultiFieldCount {

   static class Employee {
      String name;
      int age;

      Employee(String name, int age) {
         this.name = name;
         this.age = age;
      }
   }

   public static void main(String[] args) {

      List<Employee> emps = new ArrayList<Employee>();
      emps.add(new Employee("Anurag", 30));
      emps.add(new Employee("Anurag", 25));
      emps.add(new Employee("Ravi", 35));

      Collections.sort(emps, new Comparator<Employee>() {
         public int compare(Employee e1, Employee e2) {
            int nameCompare = e1.name.compareTo(e2.name);
            if (nameCompare != 0) {
               return nameCompare;
            }
            return e1.age - e2.age;
         }
      });

      System.out.println(emps.size());
   }
}
```

16 peek() logging + count()

```
import java.util.*;

public class PeekLoggingCount {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(1, 2, 3);

      long count = 0;

      for (Integer n : list) {
         System.out.println("Peek: " + n);
         count++;
      }

      System.out.println(count);
   }
}
```

---

17 peek() debugging + count()

```
import java.util.*;

public class PeekDebugCount {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(10, 20, 30);

      long count = 0;

      for (Integer n : list) {
         if (n == 20) {
            System.out.println("Debug hit");
         }
         count++;
      }

      System.out.println(count);
   }
}
```

---

18 peek() metrics + count()

```
import java.util.*;

public class PeekMetricsCount {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(5, 10, 15);

      int metricCounter = 0;
      long count = 0;

      for (Integer n : list) {
         metricCounter++;
         count++;
      }

      System.out.println("Stream count: " + count);
      System.out.println("Metric count: " + metricCounter);
   }
}
```